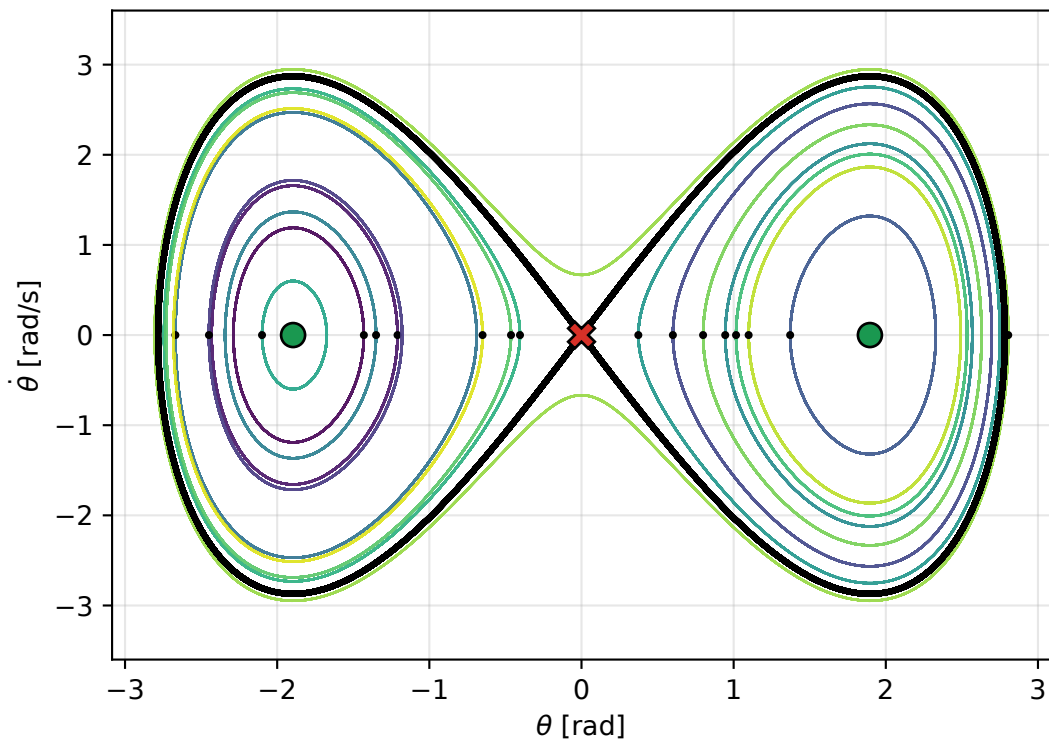


Separatrix, $k = 0.5 k_{cr}$



separatrix (simulated, $E = mgl$)

 stable (local min, $U'' > 0$)

 unstable (local max, $U'' < 0$)